

2017-Ph-H1-Q24

Section: Electricity

Topic: Current, PD, Power, Resistance

A voltage divider contains two resistors.
What is the output voltage?

 **Use Voltage Divider Rule:**

$$V_{\text{out}} = \frac{R_2}{R_1 + R_2} \times V_{\text{in}}$$

Given:

- $R_1 = 2.0 \, \Omega$,
- $R_2 = 4.0 \, \Omega$,
- $V_{\text{in}} = 12 \, \text{V}$

$$V_{\text{out}} = \frac{4}{2 + 4} \times 12 = \frac{4}{6} \times 12 = 8 \, \text{V}$$

 **Final Answer: C — 8.0 V**

 **Revision Tips:**

- Output is **across** the lower resistor in the series